

MP1: Experimental Protocols

The purpose of this project is for you create an experimental protocol for an experiment with a human and a robot. This is also an opportunity for you to get used to ROS and UR3 robot, which will be used for MP2.

You will be given a real world setup in which you (a human agent) will share a workspace with a UR3 robot. The robot needs to pause its motion when the human gets too close. More details are presented in **Real World Setup**. **Setup Instructions** and **How to Run** are provided to help you get started on the code implementation. Now, it is your job to implement the robot safety function, and then create and carry out a mini-experiment in which you test the safety of the system. Feel free to modify the code and make changes to create an interesting experiment. Summarize your protocol in your **Report**, providing the following:

- a short description of the problem statement;
- describe the safety function you implemented (safety region, stop mechanism);
- the hypothesis you would like to test and a clear statement of what the independent and dependent variables are;
- a short description of the experiment you'll use to test this hypothesis;
- a list of potential risks to the subjects and potential unintended consequences of the system;
- a set of metrics you'll be testing to verify your hypothesis and what you would expect to see;
- results from a sample experiment that you carried out;
- whether the data you collected support your hypothesis (note that we do not expect this to be statistically significant).

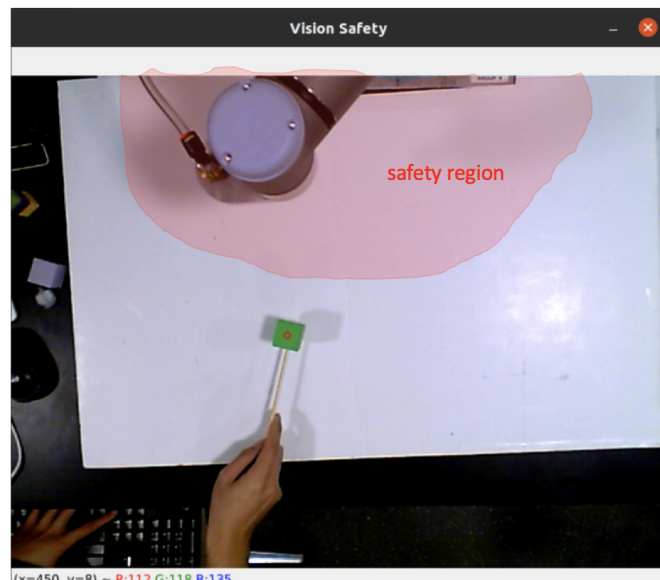
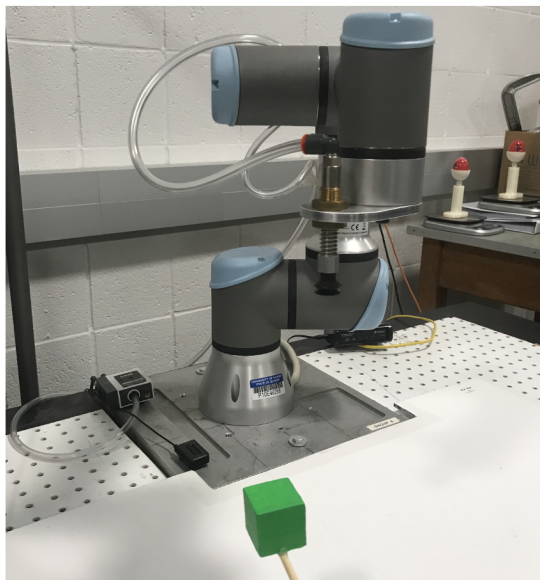
In addition to the report, you need to demonstrate your safety function to TA. You can either present a demonstration to TA during lab sections, or record both the screen and the real world and send the video to TA via email (andrems2@illinois.edu). The same **Setup Instructions** and **How to Run** will also be provided in the README-MP1.md from the zip file of code ece598hri-fa25-mps.zip.

You may submit your reports via Gradescope in groups up to 3 (fewer than 3 preferred).

We recommend you submit your MP1 report by 9/24, but will accept all MP reports until 11/7.

Real World Setup

For safety reasons, instead of putting your hand directly in the workspace, you will hold a green block on a stick to enter the robot's workspace. The green block will serve as a proxy for the human that can easily be detected. The UR3 robot is programmed to perform repetitive goal reaching motions. A top-down view camera monitors the shared space around the robot. We provide a detection algorithm for the green block, and a pause function of the robot. You will need to define a reasonable safety region around the robot, and the signal for calling the pause function of the robot to enforce safety for this shared workspace scenario. When the human agent enters the safety region, the robot motion should be suspended. The robot motion should not be resumed until the human agent exits the safety region.



Setup Instructions

1. Download `ece598hri-fa25-mps.zip`.
2. Unzip the downloaded zip file and go into the root folder `ece598hri-fa25-mps/`.
3. You will see in the root folder a catkin workspace `ws_hri/` which we will be using through the first two MPs. Go to `ws_hri/` and build the packages.

```
cd ws_hri/  
catkin_make
```

How to Run

1. Turn on UR3. Make sure Emergency button and brakes are released.
2. Open the first terminal, and run

```
cd ws_hri && source devel/setup.bash
roslaunch ur3_driver vision_driver.launch
```

2. Open the second terminal, and run

```
cd ws_hri && source devel/setup.bash
roslaunch hri mp1.py
```

The robot will start moving, and a window will pop up with a top-down view of the shared space. If you move the green block into the camera view, you should see a small red circle at the center of the green block, which indicates the green block is detected.